Sign Up

Domain Name System

What is Ip Address?

- IP Address, is a unique numerical label assigned to each server or device connected to internet.
- In simple terms, it helps one device to locate another device on the internet.

What is Domain Name?

Its a human readable/friendly address used to identify a server or device on the internet.
Like: google.com, amazon.com etc....

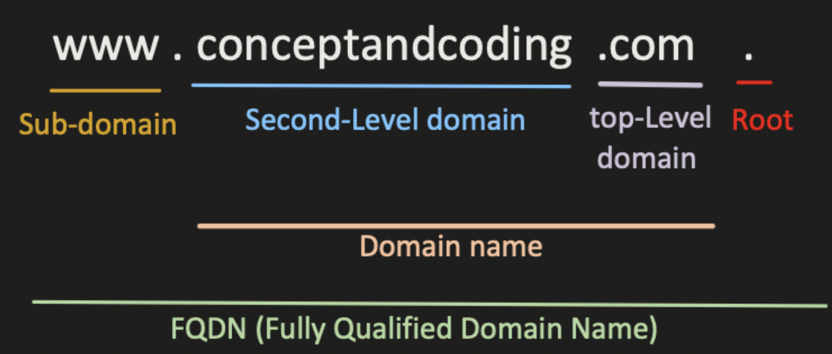
DNS:

DNS helps to translates domain name into IP Addresses.

But How?

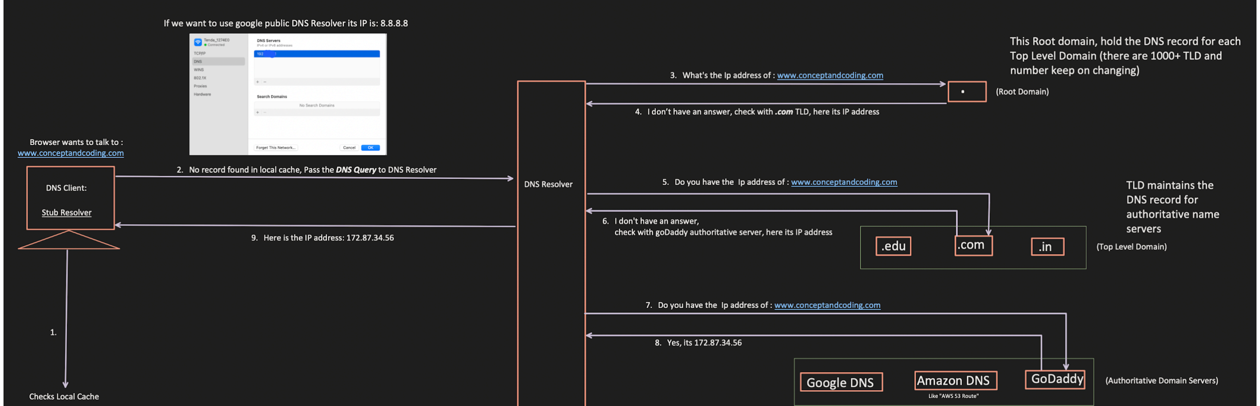
There are 2 approaches:

- Recursive
- Iterative



www . conceptandcoding .com .
Sub-domain
Second-Level domain
top-Level domain
Root
Domain name
FQDN (Fully Qualified Domain Name)

1. Recursive



If we want to use google public DNS Resolver its IP is: 8.8.8.8
Browser wants to talk to : www.conceptandcoding.com
DNS Client:
Stub Resolver
1.
Checks Local Cache
2. No record found in local cache, Pass the DNS Query to DNS Resolver
9. Here is the IP address: 172.87.34.56
DNS Resolver
3. What's the Ip address of : www.conceptandcoding.com
4. I don't have an answer, check with .com TLD, here its IP address
.
(Root Domain)
This Root domain, hold the DNS record for each Top Level Domain (there are 1000+ TLD and number keep on changing)
5. Do you have the Ip address of : www.conceptandcoding.com
6. I don't have an answer, check with goDaddy authoritative server, here its IP address
.edu
.com
.in
(Top Level Domain)
TLD maintains the DNS record for authoritative name servers
7. Do you have the Ip address of : www.conceptandcoding.com
8. Yes, its 172.87.34.56
Google DNS
Amazon DNS
Like "AWS 53 Route"
GoDaddy
(Authoritative Domain Servers)

In terminal, we can check the list of DNS which are cached by DNS Client using: ***ipconfig/displaydns***

```
C:\Users\hp>ipconfig/displaydns

Windows IP Configuration

    edgedl.me.gvt1.com
    ----------------------------------------
    Record Name . . . . . : edgedl.me.gvt1.com
    Record Type . . . . . : 1
    Time To Live  . . . . : 956
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 34.104.35.123


    c-ring-fallback.msedge.net
    ----------------------------------------
    Record Name . . . . . : c-ring-fallback.msedge.net
    Record Type . . . . . : 5
    Time To Live  . . . . : 37
    Data Length . . . . . : 8
    Section . . . . . . . : Answer
    CNAME Record  . . . . : c-9999.c-dc-msedge.net


    Record Name . . . . . : c-9999.c-dc-msedge.net
    Record Type . . . . . : 1
    Time To Live  . . . . : 37
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 13.107.12.254


    crl3.digicert.com
    ----------------------------------------
    Record Name . . . . . : crl3.digicert.com
    Record Type . . . . . : 5
    Time To Live  . . . . : 1777
    Data Length . . . . . : 8
    Section . . . . . . . : Answer
    CNAME Record  . . . . : crl.edge.digicert.com
```

We can check the Root servers list at : IANA (Internet Assigned Number Authority)
Each DNS Resolver hold this information .

1. **A-root**: Operated by Verisign, Inc.
   - Record Name: a.root-servers.net
2. **B-root**: Operated by U.S.C. Information Sciences Institute.
   - Record name: b.root-servers.net
3. **C-root**: Operated by Cogent Communications.
   - Record name: c.root-servers.net
4. **D-root**: Operated by the University of Maryland.
   - Record name: d.root-servers.net
5. **E-root**: Operated by NASA Ames Research Centre.
   - Record name: e.root-servers.net
6. **F-root**: Operated by Internet Systems Consortium (ISC).
   - Record name: f.root-servers.net
7. **G-root**: Operated by the U.S. Department of Defense (NIC).
   - Record name: g.root-servers.net
8. **H-root**: Operated by the U.S. Army Research Lab.
   - Record name: h.root-servers.net
9. **I-root**: Operated by Netnod.
   - Record name: i.root-servers.net
10. **J-root**: Operated by Verisign, Inc.
    - Record name: j.root-servers.net
11. **K-root**: Operated by RIPE NCC.
    - Record name: k.root-servers.net
12. **L-root**: Operated by ICANN.
    - Record name: l.root-servers.net
13. **M-root**: Operated by WIDE Project.
    - Record name: m.root-servers.net

One DNS Record has below main information:

Record Name: contains domain name or sub-domain name

CNAME Record:

- Canonical Name, is used to ALIAS one domain to another (subdomain only)
- example: ***www.conceptandcoding.com*** and ***blog.conceptandcoding.com*** is an alias of conceptandcoding.com

A (Host) Record : Address record maps a record name to an IP Address.

Example:

Record Name: @
Type: 1
A(Host): 194.45.32.21

Record Name: www
Type: 5
CNAME Record: conceptandcoding.com.

How TLD, knows that 'conceptandcoding.com' belongs to 'GoDaddy' authoritative server?

When user apply for any domain like 'conceptandcoding.com'

1. GoDaddy(Registrar) communicate with respective TLD registry.

2. .com Registry (Verisign maintained it) receives the request + the name servers of the GoDaddy which will be used to resolve the domain request for 'conceptandcoding.com'

What is DNS Zone?

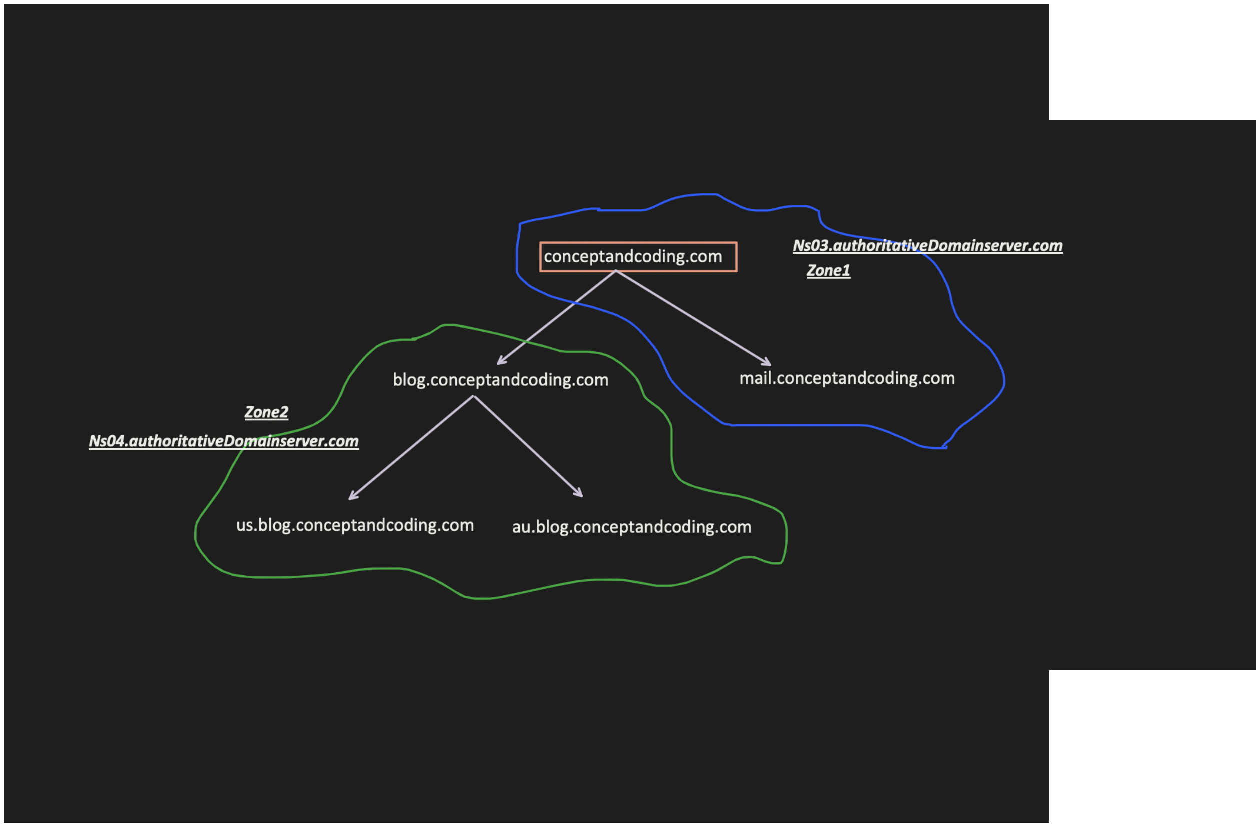
- Consider it as a separate area, which helps to provide more granular control over that.
- It also helps to offload the responsibility from the single authoritative domain server.

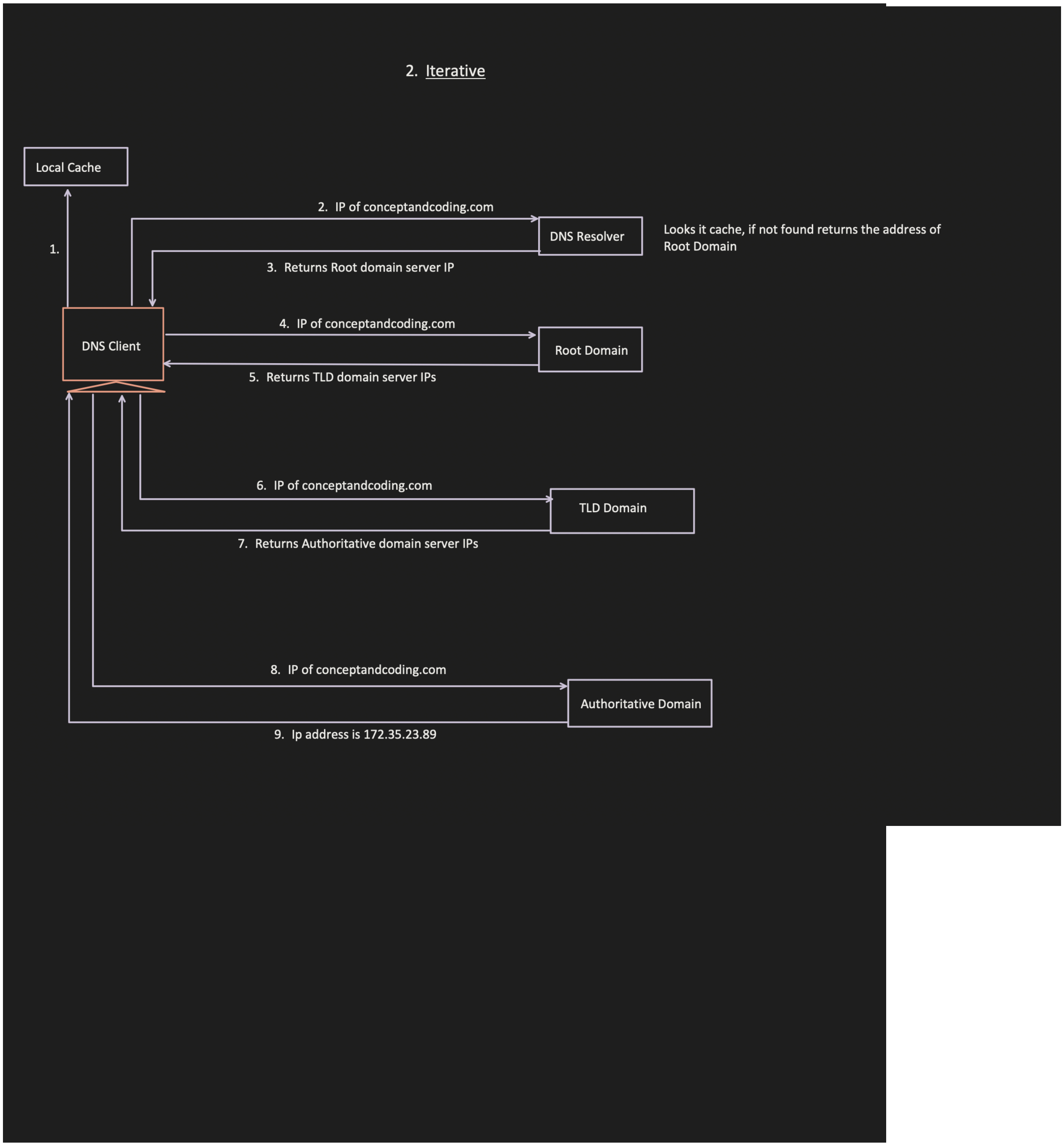
For example:

conceptandcoding.com
mail.conceptandcoding.com
blog.conceptandcoding.com
admin.conceptandcoding.com
cdn.image.media.conceptandcoding.com
systemdesign.resource.conceptandcoding.com
Etc..
Etc..

If Zone is not there, only 1 authoritative server, which hold the DNS record (Record Name, Ip Address, TTL etc..) query for all the subdomains.

But we can distribute the load, by creating more authoritative domain servers, just see how load is distributed between TLD servers '.com', '.au' likewise at authoritative level, instead of putting load on 1 server, we can distributed the traffic.


conceptandcoding.com
Ns03.authoritativeDomainserver.com
Zone1
blog.conceptandcoding.com
mail.conceptandcoding.com
Zone2
Ns04.authoritativeDomainserver.com
us.blog.conceptandcoding.com
au.blog.conceptandcoding.com


2. Iterative
Local Cache
1.
2. IP of conceptandcoding.com
DNS Resolver
Looks it cache, if not found returns the address of Root Domain
3. Returns Root domain server IP
DNS Client
4. IP of conceptandcoding.com
Root Domain
5. Returns TLD domain server IPs
6. IP of conceptandcoding.com
TLD Domain
7. Returns Authoritative domain server IPs
8. IP of conceptandcoding.com
Authoritative Domain
9. Ip address is 172.35.23.89